

The Discrete Laplacian: A Rigorous Derivation and Spectral Analysis within Axiomatic Physical Homeostasis

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Abstract

The Discrete Laplace operator, denoted as Δ_h or $\mathbf{L}$, serves as the fundamental bridge between continuum field theory and the computational substrate of the Axiomatic Physical Homeostasis (APH) framework. This paper provides a rigorous derivation of the operator from first principles, utilizing Taylor series expansions to bound truncation errors on structured lattices. We analyze the spectral properties of the operator on finite grids, establishing the connection between its eigenvectors and the coherence of the simulated state. Furthermore, we investigate the isotropy of high-order stencils (9-point vs 5-point) as a mechanism for preserving rotational invariance in the substrate. Finally, we define the strict stability conditions for entropic flux and explore the operator's role in the Helmholtz-Hodge Decomposition, the cornerstone of divergence-free fluid projection in the APH Framework.

1 Introduction

In continuum calculus, the Laplace operator $\Delta = \nabla^2 = \nabla \cdot \nabla$ measures the local divergence of the gradient of a scalar field. Within the APH framework, it is the **Operator of Equilibrium**. It dictates how energy, color, or velocity diffuses through the substrate to restore homeostasis.

To simulate these phenomena in a computational substrate—a discretized spacetime—we must project this operator onto a finite-dimensional manifold. The discrete Laplacian is not unique; its form defines the topology of the interaction graph (mesh) and the causal connectivity of the simulation. For a scalar field $u : \Omega \rightarrow \mathbb{R}$ defined on a grid $\mathcal{G}_h$, the operator Δ_h approximates the local geometric curvature of the field.

2 Formal Definition on Rectangular Lattices

Let $\Omega_h \subset \mathbb{Z}^d$ be a regular lattice acting as the discrete manifold with grid spacing h . A scalar field $u_{i,j}$ represents the state of the substrate at node $(x_i, y_j) = (ih, jh)$.

2.1 The 5-Point Stencil (Orthogonal Horizon)

Derived via the central difference approximation of the second derivative, the standard 5-point stencil in 2D represents the minimal causal horizon:

$$\Delta_h^{(5)} u_{i,j} = \frac{u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{i,j}}{h^2} \quad (1)$$

This operator is the sum of the second partial derivatives:

$$\Delta_h^{(5)} = \delta_x^2 + \delta_y^2 \quad (2)$$

where the central difference operator δ_x^2 is:

$$\delta_x^2 u_{i,j} = \frac{1}{h^2} (u_{i+1,j} - 2u_{i,j} + u_{i-1,j}) \quad (3)$$

2.2 Truncation Error Analysis

Using the Taylor series expansion for $u(x \pm h, y)$ to analyze the fidelity of the simulation:

$$u(x \pm h) = u \pm hu_x + \frac{h^2}{2}u_{xx} \pm \frac{h^3}{6}u_{xxx} + \frac{h^4}{24}u_{xxxx} + O(h^5) \quad (4)$$

Summing $u(x + h) + u(x - h)$ cancels odd derivative terms:

$$u_{i+1,j} + u_{i-1,j} = 2u_{i,j} + h^2 \frac{\partial^2 u}{\partial x^2} + \frac{h^4}{12} \frac{\partial^4 u}{\partial x^4} + O(h^6) \quad (5)$$

Substituting back into the definition of $\Delta_h^{(5)}$:

$$\Delta_h^{(5)} u = \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \frac{h^2}{12} \left(\frac{\partial^4 u}{\partial x^4} + \frac{\partial^4 u}{\partial y^4} \right) + O(h^4) \quad (6)$$

The leading error term is $O(h^2)$. However, the error term $\frac{h^2}{12}(\partial_x^4 + \partial_y^4)$ is *anisotropic*. It implies that the substrate has a preferred direction (the grid axes), which can lead to *screen door* artifacts in wave propagation.

3 Isotropic Improvement: The 9-Point Stencil

To mitigate grid artifacts and ensure rotational invariance in the simulation, we introduce the isotropic 9-point stencil. We utilize the finite difference approximations for the cross-terms and diagonals.

The “Mehrstellen” or isotropic Laplacian $\Delta_h^{(9)}$ is defined as:

$$\Delta_h^{(9)} = \frac{1}{6h^2} \begin{bmatrix} 1 & 4 & 1 \\ 4 & -20 & 4 \\ 1 & 4 & 1 \end{bmatrix} \quad (7)$$

This stencil eliminates the lowest-order anisotropic error terms, ensuring that the refractive index of the substrate is rotationally invariant. Geometrically, this expands the causal horizon to include diagonal neighbors, smoothing the propagation of flux in non-axial directions.

4 Spectral Properties and Coherence

The eigenvalues of the Laplacian determine the stiffness of the APH system. Consider the eigenvalue problem $-\Delta_h \mathbf{v} = \lambda \mathbf{v}$ on an $N \times N$ grid with Dirichlet boundary conditions ($u = 0$ on $\partial\Omega$).

The eigenfunctions are discrete sinusoids:

$$v_{k,l}(x, y) = \sin\left(\frac{k\pi x}{N+1}\right) \sin\left(\frac{l\pi y}{N+1}\right) \quad (8)$$

for $k, l = 1, \dots, N$.

The corresponding eigenvalues $\lambda_{k,l}$ for the 5-point stencil are:

$$\lambda_{k,l} = \frac{4}{h^2} \left[\sin^2\left(\frac{k\pi}{2(N+1)}\right) + \sin^2\left(\frac{l\pi}{2(N+1)}\right) \right] \quad (9)$$

In APH, these eigenvalues correspond to the **Decay Rates of Causal Structures**.

- **Low λ (Fundamental Harmonics):** These modes persist. They represent the global geometry of the setting or physical system (sky, horizon).
- **High λ (Transient Noise):** These modes decay rapidly ($e^{-\lambda t}$). They represent high-frequency shimmer or entropic noise.

5 Stability in Parabolic Systems

For the entropic flux equation $\partial_t u = \alpha \Delta u$, discretized explicitly as:

$$\frac{u^{n+1} - u^n}{\Delta t} = \alpha \Delta_h u^n \quad (10)$$

Von Neumann stability analysis requires that the amplification factor G satisfies $|G| \leq 1$. Substituting a Fourier mode $u_j^n = \xi^n e^{ikjh}$:

$$\xi = 1 - 4r \sin^2\left(\frac{kh}{2}\right) \quad (11)$$

where $r = \frac{\alpha \Delta t}{h^2}$ is the Courant-Friedrichs-Lewy (CFL) number for diffusion. Stability requires:

$$\Delta t \leq \frac{h^2}{4\alpha} \quad (12)$$

This is the **Speed Limit of Causality**. If the timestep exceeds this threshold, the substrate loses coherence, and the simulation collapses.

6 The Helmholtz-Hodge Decomposition

In fluid simulation (Navier-Stokes), the Laplacian appears in the pressure projection step. According to the Fundamental Theorem of Vector Calculus, any vector field $\mathbf{w}$ can be decomposed into:

$$\mathbf{w} = \mathbf{u}_{\text{solenoidal}} + \nabla p \quad (13)$$

where $\nabla \cdot \mathbf{u}_{\text{solenoidal}} = 0$. Taking the divergence of both sides:

$$\nabla \cdot \mathbf{w} = \nabla \cdot (\nabla p) = \Delta p \quad (14)$$

This results in the Poisson equation:

$$\Delta_h p = \nabla_h \cdot \mathbf{w} \quad (15)$$

Solving this sparse linear system $\mathbf{Lp} = \mathbf{b}$ effectively projects the divergent velocity field onto a divergence-free manifold, enforcing incompressibility in the fluid substrate.

7 The APH Interpretation: Operator Mechanics in Computational Substrate

Within the framework of APH, the Discrete Laplacian Δ_h is not merely a statistical approximation of a continuous derivative; it is the fundamental *interaction kernel* of the substrate. It defines how information propagates through the discrete manifold.

7.1 The Stencil as the Causal Horizon

In APH spacetime, causality is strictly local. A node (i, j) in the substrate cannot see the global state; it can only perceive its immediate neighbors. The stencil of the Laplacian defines the *Causal Horizon* of a single quanta of space.

$$\mathcal{H}_{\text{causal}}(i, j) = \{(x, y) \mid \mathbf{A}_{(i,j),(x,y)} \neq 0\} \quad (16)$$

where $\mathbf{A}$ is the adjacency matrix. The operation $\mathbf{L}u$ is therefore the measure of *local tension*. If $\Delta_h u_{i,j} \neq 0$, the substrate at that point is under stress relative to its neighbors, generating a restorative flux $\mathbf{J}$ proportional to the local curvature.

7.2 Self Organization as Spectral Resonance

The phenomenon of emergent, self-organizing complexity observed in simulations and in physical settings is mathematically equivalent to the excitation of the Laplacian's low-frequency eigenvectors.

Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_N\}$ be the orthonormal eigenvectors of the Laplacian matrix $\mathbf{L}$, ordered by eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$. Any state Ψ can be decomposed into this spectral basis:

$$\Psi = \sum_{k=1}^N c_k \mathbf{v}_k \quad (17)$$

In APH, the coefficients c_k represent the **coherence** of the system.

7.3 The Entropic Flux Equation

We formally define the evolution of the substrate field ϕ (representing charge, momentum, or energy) using the APH Flux Equation, which modifies the standard heat equation with a reactive source term:

$$\frac{\partial \phi}{\partial t} = \underbrace{-\mu \Delta_h \phi}_{\text{Dissipation}} + \underbrace{\mathcal{S}(\phi, t)}_{\text{Injection}} - \underbrace{(\mathbf{u} \cdot \nabla_h) \phi}_{\text{Advection}} \quad (18)$$

Here, μ is the *Viscosity of the Substrate*.

- If $\mu \rightarrow \infty$, the system freezes into a solid crystal (Laplace Equation $\Delta \phi = 0$).
- If $\mu \rightarrow 0$, the system becomes a superfluid, preserving all turbulence indefinitely.

The discrete Laplacian Δ_h in the dissipation term provides the mechanism for *Entropic Decay*. In the accompanying software implementation (“Dali”), this injection term $\mathcal{S}$ corresponds to the user-controlled spawning of stars and the procedural generation of geometry, preventing the system from relaxing into a trivial equilibrium state.

8 Conclusion

The Discrete Laplace Operator is not merely a numerical approximation; it is the topological signature of the discrete nature of the substrate of space-time itself. By rigorously adhering to the stability conditions imposed by the spectral gap and utilizing isotropic stencils, we ensure the simulation preserves physical invariants essential for the emergence of coherent structure.

References

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